

Outline of the derivation of the method of regularized Stokeslets in two dimensions based on Cortez's paper [1]

For three dimensions, refer to Cortez et al. [2]

Operators

Given a **scalar** function $p(x_1, x_2)$, define

$$\begin{aligned} \text{a) } \nabla p &= \left(\frac{\partial p}{\partial x_1}, \frac{\partial p}{\partial x_2} \right). \\ \text{b) } \Delta p &= \frac{\partial^2 p}{\partial x_1^2} + \frac{\partial^2 p}{\partial x_2^2}. \end{aligned}$$

Given a **vector** function $\mathbf{u}(x_1, x_2) = (u_1(x_1, x_2), u_2(x_1, x_2))$, define

$$\begin{aligned} \text{c) } \nabla \cdot \mathbf{u} &= \frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2}. \\ \text{d) } \Delta \mathbf{u} &= \left(\frac{\partial^2 u_1}{\partial x_1^2} + \frac{\partial^2 u_1}{\partial x_2^2}, \frac{\partial^2 u_2}{\partial x_1^2} + \frac{\partial^2 u_2}{\partial x_2^2} \right). \end{aligned}$$

Remarks

$$\begin{aligned} \text{e) } \nabla \cdot \Delta \mathbf{u} &= \Delta(\nabla \cdot \mathbf{u}) \\ \nabla \cdot \Delta \mathbf{u} &= \frac{\partial(\frac{\partial^2 u_1}{\partial x_1^2} + \frac{\partial^2 u_1}{\partial x_2^2})}{\partial x_1} + \frac{\partial(\frac{\partial^2 u_2}{\partial x_1^2} + \frac{\partial^2 u_2}{\partial x_2^2})}{\partial x_2} \\ \Delta(\nabla \cdot \mathbf{u}) &= \frac{\partial^2(\frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2})}{\partial x_1^2} + \frac{\partial^2(\frac{\partial u_1}{\partial x_1} + \frac{\partial u_2}{\partial x_2})}{\partial x_2^2} \\ \text{f) } \nabla \cdot (\nabla p) &= \nabla^2 p = \Delta p \\ \nabla \cdot (\nabla p) &= \frac{\partial}{\partial x_1} \frac{\partial p}{\partial x_1} + \frac{\partial}{\partial x_2} \frac{\partial p}{\partial x_2} \\ \Delta p &= \frac{\partial^2 p}{\partial x_1^2} + \frac{\partial^2 p}{\partial x_2^2} \end{aligned}$$

The steady Stokes equations

$$\begin{aligned} (1) \quad \mu \Delta \mathbf{u} &= \nabla p - \mathbf{F}(\mathbf{x}) \\ (2) \quad \nabla \cdot \mathbf{u} &= 0 \end{aligned}$$

where

$$(3) \quad \mathbf{F}(\mathbf{x}) = \mathbf{f}_0 \phi_\epsilon(\mathbf{x} - \mathbf{x}_0)$$

Equation (1) means that

$$\mu \left(\frac{\partial^2 u_1}{\partial x_1^2} + \frac{\partial^2 u_1}{\partial x_2^2}, \frac{\partial^2 u_2}{\partial x_1^2} + \frac{\partial^2 u_2}{\partial x_2^2} \right) = \left(\frac{\partial p}{\partial x_1}, \frac{\partial p}{\partial x_2} \right) - (F_1, F_2)$$

i.e.,

$$\begin{cases} \mu \left(\frac{\partial^2 u_1}{\partial x_1^2} + \frac{\partial^2 u_1}{\partial x_2^2} \right) = \frac{\partial p}{\partial x_1} - F_1 \\ \mu \left(\frac{\partial^2 u_2}{\partial x_1^2} + \frac{\partial^2 u_2}{\partial x_2^2} \right) = \frac{\partial p}{\partial x_2} - F_2 \end{cases}$$

where

$$\mathbf{F}(\mathbf{x}) = (F_1, F_2) = (f_{01}, f_{02}) \phi_\epsilon(\mathbf{x} - \mathbf{x}_0)$$

Derive the solutions p and $\mathbf{u}$ of Equations (1), (2) and (3)

*** Find p :

$$\begin{aligned} \nabla \cdot (\mu \Delta \mathbf{u}) &= \nabla \cdot (\nabla p - \mathbf{F}(\mathbf{x})) \\ \rightarrow \mu \nabla \cdot (\Delta \mathbf{u}) &= \nabla \cdot (\nabla p) - \nabla \cdot \mathbf{F} \\ \rightarrow \mu \Delta(\nabla \cdot \mathbf{u}) &= \Delta p - \nabla \cdot \mathbf{F} \\ \rightarrow \Delta p &= \nabla \cdot \mathbf{F} \\ \rightarrow \Delta p &= \nabla \cdot (\mathbf{f}_0 \phi_\epsilon) \\ \rightarrow \Delta p &= \mathbf{f}_0 \cdot \nabla \phi_\epsilon \end{aligned}$$

Since $\Delta G_\epsilon = \phi_\epsilon$, $\nabla \phi_\epsilon = \nabla(\Delta G_\epsilon)$. So

$$\begin{aligned}\rightarrow \Delta p &= \mathbf{f}_0 \cdot \nabla \phi_\epsilon \\ \rightarrow \Delta p &= \mathbf{f}_0 \cdot \nabla(\Delta G_\epsilon) \\ \rightarrow p &= \mathbf{f}_0 \cdot \nabla G_\epsilon\end{aligned}$$

*** Find $\mathbf{u}$:

Since $p = \mathbf{f}_0 \cdot \nabla G_\epsilon$, $\nabla p = \nabla(\mathbf{f}_0 \cdot \nabla G_\epsilon) = (\mathbf{f}_0 \cdot \nabla) \nabla G_\epsilon$.

$$\begin{aligned}\mu \Delta \mathbf{u} &= \nabla p - \mathbf{F}(\mathbf{x}) \\ \rightarrow \mu \Delta \mathbf{u} &= (\mathbf{f}_0 \cdot \nabla) \nabla G_\epsilon - \mathbf{f}_0 \phi_\epsilon\end{aligned}$$

Since $\Delta B_\epsilon = G_\epsilon$, $\nabla G_\epsilon = \nabla(\Delta B_\epsilon)$.

$$\begin{aligned}\mu \Delta \mathbf{u} &= (\mathbf{f}_0 \cdot \nabla) \nabla G_\epsilon - \mathbf{f}_0 \phi_\epsilon \\ \rightarrow \mu \Delta \mathbf{u} &= (\mathbf{f}_0 \cdot \nabla) \nabla(\Delta B_\epsilon) - \mathbf{f}_0 \Delta B_\epsilon \\ \rightarrow \mu \mathbf{u} &= (\mathbf{f}_0 \cdot \nabla) \nabla B_\epsilon - \mathbf{f}_0 G_\epsilon\end{aligned}$$

This is a *regularized* Stokeslet velocity.

Discrete versions of p and $\mathbf{u}$

If there are N forces $\mathbf{f}_k$, centered at points $\mathbf{x}_k$, the pressure and velocity are given by

$$(6) \quad p(\mathbf{x}) = \sum_{k=1}^N \mathbf{f}_k \cdot \nabla G_\epsilon(\mathbf{x} - \mathbf{x}_k),$$

$$(7) \quad \mathbf{u}(\mathbf{x}) = \mathbf{U}_o + \frac{1}{\mu} \sum_{k=1}^N \{(\mathbf{f}_k \cdot \nabla) \nabla B_\epsilon(\mathbf{x} - \mathbf{x}_k) - \mathbf{f}_k G_\epsilon(\mathbf{x} - \mathbf{x}_k)\},$$

where $\mathbf{U}_o$ will be determined later.

Let $r = |\mathbf{x}|$, we have the following transformations:

$$\begin{aligned}\nabla G_\epsilon &= G'_\epsilon(r) \frac{\mathbf{x}}{r} \\ \nabla B_\epsilon &= B'_\epsilon(r) \frac{\mathbf{x}}{r} \\ (\mathbf{f} \cdot \nabla) \nabla B_\epsilon &= \mathbf{f} \frac{B'_\epsilon(r)}{r} + (\mathbf{f} \cdot \mathbf{x}) \mathbf{x} \left[\frac{r B''_\epsilon(r) - B'_\epsilon(r)}{r^3} \right]\end{aligned}$$

Let $r_k = |\mathbf{x} - \mathbf{x}_k|$, and use the above transformations to change Equations (6) and (7) to produce the following equations:

$$(8) \quad p(\mathbf{x}) = \sum_{k=1}^N [\mathbf{f}_k \cdot (\mathbf{x} - \mathbf{x}_k)] \left[\frac{G'_\epsilon(r_k)}{r_k} \right],$$

$$(9) \quad \begin{aligned}\mathbf{u}(\mathbf{x}) &= \mathbf{U}_o + \frac{1}{\mu} \sum_{k=1}^N \left\{ \mathbf{f}_k \left[\frac{B'_\epsilon(r_k)}{r_k} - G_\epsilon(r_k) \right] \right. \\ &\quad \left. + [\mathbf{f}_k \cdot (\mathbf{x} - \mathbf{x}_k)] (\mathbf{x} - \mathbf{x}_k) \left[\frac{r_k B''_\epsilon(r_k) - B'_\epsilon(r_k)}{r_k^3} \right] \right\}\end{aligned}$$

Derive the solutions p and $\mathbf{u}$ based on a given blob function in two dimensions
 Given the following blob

$$\phi_\epsilon(\mathbf{x}) = \frac{3\epsilon^3}{2\pi(|\mathbf{x}|^2 + \epsilon^2)^{5/2}}$$

*** Find G_ϵ :

Since $\triangle G_\epsilon = \frac{1}{r}[rG'_\epsilon(r)]'$ and $\triangle G_\epsilon = \phi_\epsilon$, we have

$$\begin{aligned} \frac{1}{r}[rG'_\epsilon(r)]' &= \phi_\epsilon(r) \\ \rightarrow [rG'_\epsilon(r)]' &= r\phi_\epsilon(r) \\ \rightarrow rG'_\epsilon(r) &= \int_0^r R\phi_\epsilon(R)dR = I_1 \\ \rightarrow G'_\epsilon(r) &= \frac{1}{r}I_1 = I_2 \\ \rightarrow G_\epsilon(r) &= \int I_2 dr \end{aligned}$$

In this case, we get

$$G_\epsilon(r) = \frac{1}{2\pi} \left[\ln \left(\sqrt{r^2 + \epsilon^2} + \epsilon \right) - \frac{\epsilon}{\sqrt{r^2 + \epsilon^2}} \right]$$

*** Find B'_ϵ :

Since $\triangle B_\epsilon = \frac{1}{r}[rB'_\epsilon(r)]'$ and $\triangle B_\epsilon = C_\epsilon$, we have

$$\begin{aligned} \frac{1}{r}[rB'_\epsilon(r)]' &= G_\epsilon(r) \\ \rightarrow [rB'_\epsilon(r)]' &= rG_\epsilon(r) \\ \rightarrow rB'_\epsilon(r) &= \int_0^r RG_\epsilon(R)dR = I_3 \\ \rightarrow B'_\epsilon(r) &= \frac{1}{r}I_3 = I_4 \end{aligned}$$

In this case, we get

$$B'_\epsilon(r) = \frac{1}{8\pi} \left[2r \ln \left(\sqrt{r^2 + \epsilon^2} + \epsilon \right) - r - \frac{2r\epsilon}{\sqrt{r^2 + \epsilon^2} + \epsilon} \right]$$

*** Find p and $\mathbf{u}$ from substituting the resulting G_ϵ and B_ϵ into Equations (8) and (9):

$$(10) \quad p(\mathbf{x}) = \sum_{k=1}^N \frac{1}{2\pi} [\mathbf{f}_k \cdot (\mathbf{x} - \mathbf{x}_k)] \left[\frac{r_k^2 + 2\epsilon^2 + \epsilon\sqrt{r_k^2 + \epsilon^2}}{\left(\sqrt{r_k^2 + \epsilon^2} + \epsilon \right) (r_k^2 + \epsilon^2)^{3/2}} \right],$$

$$\begin{aligned} \mathbf{u}(\mathbf{x}) &= \sum_{k=1}^N \frac{-\mathbf{f}_k}{4\pi\mu} \left[\ln \left(\sqrt{r_k^2 + \epsilon^2} + \epsilon \right) - \frac{\epsilon \left(\sqrt{r_k^2 + \epsilon^2} + 2\epsilon \right)}{\left(\sqrt{r_k^2 + \epsilon^2} + \epsilon \right) \sqrt{r_k^2 + \epsilon^2}} \right] \\ (11) \quad &+ \frac{1}{4\pi\mu} [\mathbf{f}_k \cdot (\mathbf{x} - \mathbf{x}_k)] (\mathbf{x} - \mathbf{x}_k) \left[\frac{\sqrt{r_k^2 + \epsilon^2} + 2\epsilon}{\left(\sqrt{r_k^2 + \epsilon^2} + \epsilon \right)^2 \sqrt{r_k^2 + \epsilon^2}} \right] \end{aligned}$$

where $r_k = |\mathbf{x} - \mathbf{x}_k|$.

Notice that $U_o = \sum_k \frac{\mathbf{f}_k}{8\pi\mu}$ in this case.

References

- [1] R. Cortez, "The Method of Regularized Stokeslets," SIAM Journal on Scientific Computing, 23(4) (2001): 1204 - 1225
- [2] R. Cortez, L. Fauci, and A. Medovikov, "The method of regularized Stokeslets in three dimensions: analysis, validation, and application to helical swimming," Physics of Fluids, 17 (2005): 1 - 14

Homework assignment (due in class on September 23)

Immerse a line segment from 0 to 1 on the x -axis into a fluid of viscosity 1. Use N points to divide the line segment into equal sub-intervals of length ds and impose a point-force $(0, 1)$ at each point.

*** Question 1: Given $N = 9$ and $\epsilon = ds$, find the pressure and velocity at

- a) $(0.5, 0)$. Draw the resulting velocity vector.
- b) each of the N points. Draw the resulting velocity vectors.
- c) every point on a grid $[-2, 3] \times [-2, 2]$. Draw the velocity field and streamline.

*** Question 2: Find the pressure and velocity at $(0.5, 0)$ for different values of N where $N \in \{3, 5, 9, 17, 33\}$ and $\epsilon = ds$. What do the pressure and velocity at this point converge to?

*** Question 3: Find the pressure and velocity at $(0.5, 0)$ for $N = 33$ and $\epsilon \in \{0.1ds, 0.2ds, \dots, 1.0ds, 1.1ds, \dots, 1.5ds\}$. Draw the plots of pressure and velocity magnitude with respect to ϵ ?